

Coble type hypersurfaces and hyperkähler fourfolds

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Goal

Provide new examples of locally complete families of *hyperkähler fourfolds* X of $K3^{[2]}$ -type by describing X as the singular locus of a hypersurface Y , which we call a *Coble type hypersurface*.

Coble type hypersurfaces

Definition

Let $X \subset \mathbf{P}^N$ be a smooth projective variety. We say that X *admits a Coble type hypersurface* $Y = V(f_d)$ of degree d , if scheme-theoretically

$$X = \text{Sing}(Y).$$

Examples (Fano)

A silly example: for a smooth quadric $Q = V(q) \subset \mathbf{P}^N$,

$$Q = \text{Sing}(V(q^2)),$$

for the non-reduced quartic hypersurface $V(q^2)$.

Examples (Fano)

The four Severi varieties $\mathbf{S}_k$ of dimension $2k$ in $\mathbf{P}^{3k+2}$ for $k \in \{1, 2, 4, 8\}$, each satisfies

$$\mathbf{S}_k = \text{Sing}(\mathbf{Y}_k),$$

where $\mathbf{Y}_k$ is the (defected) secant variety, which is a cubic hypersurface.

- $\mathbf{S}_1 := v_2(\mathbf{P}^2) \subset \mathbf{P}^5$ Veronese surface and the discriminant cubic $\mathbf{Y}_1$.
- $\mathbf{S}_2 := \mathbf{P}^2 \times \mathbf{P}^2 \subset \mathbf{P}^8$ Segre fourfold and the determinantal cubic $\mathbf{Y}_2$.
- $\mathbf{S}_4 := \text{Gr}(2, 6) \subset \mathbf{P}^{14}$ Grassmannian and the Pfaffian cubic $\mathbf{Y}_4$.
- $\mathbf{S}_8 := E_6/P_1 \subset \mathbf{P}^{26}$ the *Cayley plane* and the *Cartan cubic* $\mathbf{Y}_8$.

The 5-dimensional moduli space $\mathcal{M}_1^8 := (\mathbf{P}^1)^8 // \text{SL}(2) \subset \mathbf{P}^{13}$ satisfies

$$\mathcal{M}_1^8 = \text{Sing}(Y)$$

for a cubic hypersurface $Y \subset \mathbf{P}^{13}$. $\mathcal{M}_1^8$ is singular at 35 points.

Examples (K-trivial)

- An elliptic quartic E in $\mathbf{P}^3$ is the complete intersection of two quadrics. A (decomposable) quartic Y in $\mathrm{Sym}^2 \langle q_1, q_2 \rangle$ satisfies

$$E = \mathrm{Sing}(Y).$$

We get a net of such quartics; along a conic in the net we get quartics that degenerate to q^2 .

- An elliptic sextic E in $\mathbf{P}^5$ can be realized as a codim. 3 linear section of $\mathbf{S}_2 = \mathbf{P}^2 \times \mathbf{P}^2 = \mathrm{Sing}(\mathbf{Y}_2) \subset \mathbf{P}^8$ so

$$E = \mathrm{Sing}(\mathbf{Y}_2 \cap \mathbf{P}^5).$$

We get a pencil of such cubics; four of them are isomorphic to the discriminant cubic $\mathbf{Y}_1$ singular along the Veronese surface $\mathbf{S}_1$.

More generally,

- For a complete intersection of $n + 1$ quadrics, a general quartic in $\mathrm{Sym}^2 \langle q_i \rangle$ is a Coble type quartic.
- When X is a Severi variety $\mathbf{S}_k$ or $\mathcal{M}_1^8$, a suitable linear section $X \cap \mathbf{P}^N$ is K-trivial and admits a Coble type cubic $Y \cap \mathbf{P}^N$.

Examples (K-trivial)

The Coble quartic and cubic for general principally polarized abelian varieties.

- $g = 3$: for $A = JC$ the Jacobian of a non-hyperelliptic genus-3 curve C , the Kummer threefold $K(A) := A/\{\pm 1\} \subset \mathbf{P}^7 = |2\Theta|^\vee$ is the singular locus of a unique quartic hypersurface $Y = V(f_4)$

$$K(A) = \text{Sing}(Y).$$

When C becomes hyperelliptic, the quartic degenerates to q^2 .

- $g = 2$: for $A = JC$ the Jacobian of a genus-2 curve, the embedded $A \subset \mathbf{P}^8 = |3\Theta|^\vee$ is the singular locus of a unique cubic hypersurface $Y = V(f_3)$

$$A = \text{Sing}(Y).$$

When $A = E_1 \times E_2$ becomes decomposable, the cubic degenerates to the determinantal cubic $\mathbf{Y}_2$ singular along the Segre $\mathbf{S}_2 = \mathbf{P}^2 \times \mathbf{P}^2$.

Remarks on the (classical) Coble hypersurfaces

Proving the existence of a hypersurface Y with $X \subset \text{Sing}(Y)$ is not hard (*Coble 1917; Beauville (2003) has an elegant argument using the representation theory of the Heisenberg group*).

However showing that X is exactly $\text{Sing}(Y)$ seems harder (*cubic case by Barth (1995) and quartic case by Laszlo (1996)*).

Nowadays, we have a more unified framework using the (orbital) degeneracy locus description of certain tensor objects ($\omega \in \bigwedge^4 \mathbb{C}^8$ and $\omega \in \bigwedge^3 \mathbb{C}^9$). These also exhibit very beautiful relations with moduli spaces of vector bundles on curves of genus 3 and 2 (*Narasimhan–Ramanan 1987, Gruson–Sam–Weyman 2013, Benedetti–Bolognesi–Faenzi–Manivel 2024, 2025, 2026*).

A condition on dimension and degree

Proposition

Let $X \subset \mathbf{P}^N$ be a smooth projective variety of dimension n . If X admits a Coble type hypersurface Y of degree d , then

$$2K_X = (d(N - n) - 2(N + 1))H.$$

Idea of proof: when X is the singular locus of the hypersurface Y , we obtain a smooth quadric bundle structure

$$q: \operatorname{Sym}^2 \mathcal{N}_{X/\mathbf{P}} \longrightarrow \mathcal{O}_X(d).$$

So we need its discriminant $\operatorname{disc}(q)$ to be a section of $\mathcal{O}_X$.

In particular, if X is K-trivial, then

$$d(N - n) - 2(N + 1) = 0 \quad \text{or equivalently} \quad N = n + \frac{2(n+1)}{d-2}.$$

Hence Coble type quartics only occur in $\mathbf{P}^{2n+1}$ and cubics only in $\mathbf{P}^{3n+2}$.

A condition on dimension and degree

n	quartic	example	cubic	example
elliptic curve	$\mathbf{P}^3$	(2^2) -c.i.	$\mathbf{P}^5$	$\mathbf{S}_2 \cap \mathbf{P}^5$
surface	$\mathbf{P}^5$	(2^3) -c.i.=K3	$\mathbf{P}^8$	$\mathbf{S}_4 \cap \mathbf{P}^8 = \mathbf{K3}$ A
threefold	$\mathbf{P}^7$	(2^4) -c.i.=CY ₃ K(A)	$\mathbf{P}^{11}$	$\mathcal{M}_1^8 \cap \mathbf{P}^{11} = \text{CY}_3$
fourfold	$\mathbf{P}^9$	(2^5) -c.i.=CY ₄ HK	$\mathbf{P}^{14}$	$\mathbf{S}_8 \cap \mathbf{P}^{14} = \text{CY}_4$ HK

Hyperkähler fourfolds

Introduction

A *hyperkähler manifold* is a simply connected compact Kähler manifold X such that

$$H^{2,0}(X) = H^0(X, \Omega_X^2) = \mathbb{C}\sigma$$

is generated by a nowhere vanishing holomorphic 2-form σ . Also called an *irreducible holomorphic symplectic* manifold.

X is necessarily of even (complex) dimension and has trivial canonical bundle. In dimension 2, these are exactly the K3 surfaces.

$$\begin{array}{ccccc}
 & & 1 & & \\
 & 0 & & 0 & \\
 1 & & 20 & & 1 \\
 & 0 & & 0 & \\
 & & 1 & &
 \end{array}
 \qquad
 \begin{array}{ccccccc}
 & & & & 1 & & \\
 & & & 0 & & 0 & \\
 & & 1 & & * & & 1 \\
 & 0 & & * & & * & 0 \\
 1 & & * & & * & & * & 1
 \end{array}$$

Hyperkähler manifolds are one of the three archetypical compact Kähler manifolds with trivial canonical bundle, with the other two being complex tori (abelian varieties) and strict Calabi–Yau.

Deformation types

We have the following known deformation types.

- K3 surface (*20-dim. family $\Rightarrow$ 19-dim. projective families*).
- The $K3^{[n]}$ -type: the Hilbert scheme $S^{[n]}$ of n points on a K3 surface S and its deformations (*20-dim. projective families*).
- The Kum_n -type (*4-dim. projective families*).
- The O'Grady examples OG6 and OG10.

Projective models (dim. 2)

A general K3 surface $(S, H) \subset \mathbf{P}^g$ of genus g (H is a primitive polarization with $H^2 = 2g - 2$) is described by Mukai in the following cases (each giving a 19-dim. family).

- $g = 2$: $S \xrightarrow{2:1} \mathbf{P}^2$ is a double cover branched along a smooth sextic.
- $g \in \{3, 4, 5\}$: S is a (4) , $(2, 3)$, $(2, 2, 2)$ -complete intersection in $\mathbf{P}^3$, $\mathbf{P}^4$, $\mathbf{P}^5$, respectively.
- $g = 6$: S is a $(1, 1, 1, 2)$ -complete intersection in $\mathrm{Gr}(2, 5) \subset \mathbf{P}^9$.
- **$g = 7$: S is a $\mathbf{P}^7$ -linear section of $\mathrm{OGr}_+(5, 10) \subset \mathbf{P}^{15}$.**
- **$g = 8$: S is a $\mathbf{P}^8$ -linear section of $\mathrm{Gr}(2, 6) \subset \mathbf{P}^{14}$.**
- $g = 9$: S is a $\mathbf{P}^9$ -linear section of $\mathrm{LGr}(3, 6) \subset \mathbf{P}^{13}$.
- $g = 10$: S is a $\mathbf{P}^{10}$ -linear section of $G_2 \subset \mathbf{P}^{13}$.
- $g \in \{11, 12, 13, 16, 18, 20\}$: explicit descriptions.

Projective models (higher dimension)

First of all, for (X, H) a polarized hyperkähler manifold, to characterize H we need at least two invariants.

- $q(H)$: the *square*, where q is the *Beauville–Bogomolov–Fujiki form* on $H^2(X, \mathbf{Z})$.
- $\text{div}(H)$: the *divisibility*, the positive generator γ of the subgroup $q(H, H^2(X, \mathbf{Z})) = \gamma\mathbf{Z}$ of $\mathbf{Z}$.

For K3 surface, we have $q(H) = H^2 = 2g - 2$ while $\text{div}(H)$ is always 1 since $H^2(K3, \mathbf{Z})$ is unimodular.

Projective models (higher dimension)

For $K3^{[2]}$ -type, denote by $\mathcal{M}_{2d}^{(\gamma)}$ the moduli space of (X, H) with square $2d$ and divisibility γ .

It is non-empty (and irreducible of dimension 20) in the following case, .

- $\gamma = 1$, all $2d \geq 2$ ($\mathcal{M}_2^{(1)} \mathcal{M}_4^{(1)} \mathcal{M}_6^{(1)} \dots$).
- $\gamma = 2$, $2d \equiv 6 \pmod{8}$ ($\mathcal{M}_6^{(2)} \mathcal{M}_{14}^{(2)} \mathcal{M}_{22}^{(2)} \dots$).

Projective models (higher dimension)

The description of a locally complete family is known only in the following cases.

- $\mathcal{M}_2^{(1)}$: (O'Grady 2006) $X \xrightarrow{2:1} Y_6 \subset \mathbf{P}^5$ is a *double EPW sextic*.
- $\mathcal{M}_6^{(2)}$: (Beauville–Donagi 1985) $X \subset \mathbf{P}^{14}$ is the *variety of lines* of a cubic fourfold $Y \subset \mathbf{P}^5$.
- $\mathcal{M}_{22}^{(2)}$: (Debarre–Voisin 2010) $X \subset \mathbf{P}^{90} \cap \mathrm{Gr}(6, 10)$.
- $\mathcal{M}_{38}^{(2)}$: (Iliev–Ranestad 2001) $X \subset \mathbf{P}^{230}$.

In higher dimensions, two more examples are known.

- $\mathcal{M}_{\mathrm{K3}^{[4]}, 2}^{(2)}$ (Lehn–Lehn–Sorger–van Straten 2017).
- $\mathcal{M}_{\mathrm{K3}^{[3]}, 4}^{(2)}$ (Iliev–Kapustka–Kapustka–Ranestad 2019).

Main result

Theorem (Piroddi–Ríos Ortiz–Rojas–S.)

We have the following descriptions for the two locally complete families (note that H is very ample for a general member).

- *A general $(X, H) \in \mathcal{M}_4^{(1)}$ in $\mathbf{P}^9$ admits a unique Coble type quartic $Y = V(f_4)$*

$$X = \text{Sing}(Y).$$

- *A general $(X, H) \in \mathcal{M}_6^{(1)}$ in $\mathbf{P}^{14}$ admits a unique Coble type cubic $Y = V(f_3)$*

$$X = \text{Sing}(Y).$$

The starting point of the main results is the following (*recall how in other examples of Coble type hypersurfaces, the quartic / cubic can degenerate to a double quadric / a secant cubic $\mathbf{Y}_k$*).

Theorem (Benedetti, Ríos Ortiz–Rojas–S.)

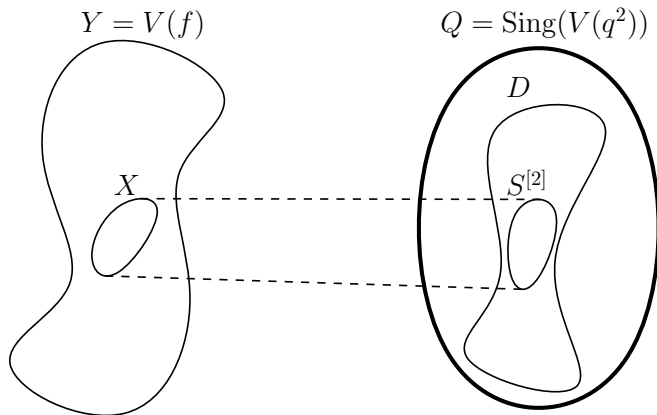
We have the following descriptions for the Hilbert squares.

- *For the fourfold $(X, H) = (S^{[2]}, L - 2\delta) \in \mathcal{M}_4^{(1)}$, where (S, L) is a general K3 surface of genus 7, X is contained in a quadric Q as the singular locus of a unique quartic $D \subset Q$*

$$X = \text{Sing}(D) \hookrightarrow Q = \text{Sing}(V(q^2)).$$

- *For the fourfold $(X, H) = (S^{[2]}, L - 2\delta) \in \mathcal{M}_6^{(1)}$, where (S, L) is a general K3 surface of genus 8, X is contained in a Grassmannian $\text{Gr}(2, 6)$ as the singular locus of a unique cubic $D \subset \text{Gr}(2, 6)$*

$$X = \text{Sing}(D) \hookrightarrow \text{Gr}(2, 6) = \mathbf{S}_4 = \text{Sing}(\mathbf{Y}_4).$$



In the case of square 4, when X specializes to $S^{[2]}$,
 $Y = V(f)$ degenerates to $V(q^2)$.

When a general $(X, H) \in \mathcal{M}_4^{(1)}$ deforms to $(S^{[2]}, L - 2\delta)$, the Coble type quartic $Y = V(f)$ degenerates to the distinguished quartic $V(q^2)$.

The direction in which f approaches q^2 is important.

If we consider a first-order deformation of $S^{[2]}$ such that the corresponding Coble quartics are given by

$$f_\varepsilon = q^2 + \varepsilon \cdot f',$$

then the equation f' cuts Q along a quartic hypersurface D , which is precisely the one appearing in

$$S^{[2]} = \text{Sing}(D) \hookrightarrow Q = \text{Sing}(V(q^2)).$$

The situation is very similar for $\mathcal{M}_6^{(1)}$.

Homogeneous ideal

We briefly comment on the homogeneous ideal of X . Note that by the Jacobian criterion, the equations $\frac{\partial f}{\partial x_i}$ lie in the ideal. But do they generate it?

- The homogeneous ideal of a general $(X, H) \in \mathcal{M}_4^{(1)}$ is generated by 10 cubics $\frac{\partial f}{\partial x_i}$ and $20 = \dim \mathcal{M}_4^{(1)}$ more quartics.
- The homogeneous ideal of a general $(X, H) \in \mathcal{M}_6^{(1)}$ is generated by 15 quadrics $\frac{\partial f}{\partial x_i}$ and $20 = \dim \mathcal{M}_6^{(1)}$ more cubics.
- The homogeneous ideal of a general $K(A) \subset \mathbf{P}^7$ is generated by 8 cubics $\frac{\partial f}{\partial x_i}$ and $6 = \dim \mathcal{A}_3$ more quartics.
- The homogeneous ideal of a general $A \subset \mathbf{P}^8$ is generated by 9 quadrics $\frac{\partial f}{\partial x_i}$ and $3 = \dim \mathcal{A}_2$ more cubics.

Homogeneous ideal

If we consider X admitting a Coble type hypersurface $Y = V(f)$, then for a (non-isotrivial) first-order deformation

$$f_\varepsilon = f + \varepsilon \cdot f',$$

the equation f' provides a degree- d equation containing X that cannot be generated by the partial derivatives of f .

Namely, if $f' = \sum_i \ell_i \cdot \frac{\partial f}{\partial x_i}$ where $\ell(x) = (\ell_0(x), \dots, \ell_N(x))$ are linear forms, then we have

$$f(x) + \varepsilon \cdot f'(x) = f(x + \varepsilon \cdot \ell(x)).$$

So after taking the first-order linear transformation $x \mapsto x + \varepsilon \cdot \ell(x)$, the deformation becomes trivial.

Moduli space

We can define (by taking closure) the locus of Coble type quartics

$$\Sigma_{\text{Coble4}}^{\dim 119} \subset |\mathcal{O}_{\mathbf{P}^9}(4)|^{\dim 714}$$

as well as the corresponding moduli space $\mathcal{M}_{\text{Coble4}} := \Sigma_{\text{Coble4}} // \text{SL}(10)$

$$\mathcal{M}_{\text{Coble4}}^{\dim 20} \subset |\mathcal{O}_{\mathbf{P}^9}(4)| // \text{SL}(10)^{\dim 615}.$$

(Similarly for $\mathcal{M}_6^{(1)}$, we have $\Sigma_{\text{Coble3}}^{\dim 244} \subset |\mathcal{O}_{\mathbf{P}^{14}}(3)|^{\dim 679}$ and $\mathcal{M}_{\text{Coble3}}^{\dim 20} \subset |\mathcal{O}_{\mathbf{P}^{14}}(3)| // \text{SL}(15)^{\dim 455}$.)

However the construction is implicit.

Problem

How can we characterize Σ_{Coble} and $\mathcal{M}_{\text{Coble}}$?

Since the codimension is huge, we are unlikely to find explicit equations / invariants. Can we instead find a (unirational) parametrization?

Moduli space

We focus on the case of square 4.

We have the following birational maps

$$\mathcal{M}_{\text{Coble4}} \xrightarrow{\mathfrak{m}} \mathcal{M}_4^{(1)} \xhookrightarrow{\mathfrak{p}} \mathcal{P}_4^{(1)},$$

where $\mathcal{M}_4^{(1)}$ is embedded inside the period domain $\mathcal{P}_4^{(1)}$ via the period map $\mathfrak{p}$ by the global Torelli theorem.

The complement of the image of $\mathfrak{p}$ is the union of three Heegner divisors

$$\mathfrak{p}: \mathcal{M}_4^{(1)} \xrightarrow{\sim} \mathcal{P}_4^{(1)} \setminus (\mathcal{D}_4 \cup \mathcal{D}_{16} \cup \mathcal{D}_{20}).$$

Heegner divisors parametrize members with Picard rank ≥ 2 .

The divisor $\mathcal{D}_{2d}$ is non-empty if and only if $2d \equiv 0, 2, 4, 6, 8, 12 \pmod{16}$ and it is then irreducible.

Moduli space

The point $[q^2]$ lies in $\mathcal{M}_{\text{Coble4}}$ (since it's a degeneration of Coble type quartics) but it corresponds to an entire Heegner divisor $\mathcal{D}_{12}$ parametrizing $(S^{[2]}, L - 2\delta)$ (the Néron–Severi lattice has Gram matrix $\begin{pmatrix} 12 & 0 \\ 0 & -2 \end{pmatrix}$).

Geometrically, we are blowing up the point $[q^2]$ in $\mathcal{M}_{\text{Coble4}}$ (recall the description with the first-order deformation) and $\mathcal{D}_{12}$ is the image of the exceptional divisor. This is a so-called *Hassett–Looijenga–Shah (HLS)* divisor.

Expectation. $\mathcal{D}_2, \mathcal{D}_4, \mathcal{D}_6, \mathcal{D}_8$, and $\mathcal{D}_{12}$ are all HLS divisors, as we have found the candidates for the limit Coble type quartic (the analogue of q^2) in these cases.

Moduli space

Conjecture

There is an open locus $\mathcal{M}_{\text{Coble4}}^{\circ} \subset \mathcal{M}_{\text{Coble4}}$ parametrizing Coble type quartics (whose singular locus is a smooth hyperkähler fourfold). We have an embedding

$$\mathfrak{m}: \mathcal{M}_{\text{Coble4}}^{\circ} \hookrightarrow \mathcal{M}_4^{(1)},$$

whose complement is the Heegner divisors $\mathcal{D}_2, \mathcal{D}_6, \mathcal{D}_8$, and $\mathcal{D}_{12}$. (Recall that the complement of $\mathcal{M}_4^{(1)}$ in $\mathcal{P}_4^{(1)}$ is $\mathcal{D}_4, \mathcal{D}_{16}, \mathcal{D}_{20}$.)

This is based on the picture for cubic fourfolds and their variety of lines (Voisin, Hassett, Looijenga, Laza, ...): in that setting, we have

$$\mathcal{M}_{\text{cubic}}^{\circ} \xhookrightarrow{\mathfrak{m}} \mathcal{M}_6^{(2)} \xhookrightarrow{\mathfrak{p}} \mathcal{P}_6^{(2)},$$

where the complement of $\mathfrak{m}$ is one Heegner divisor $\mathcal{D}_2$ (*blowing up the class of the discriminant cubic $\mathbf{Y}_1$*), and the complement of $\mathfrak{p}$ is one Heegner divisor $\mathcal{D}_6$ (*nodal cubics*).

Thank you!

Joyeux anniversaire Laurent !

Hodge structure (known examples)

Cubic fourfold $Y \subset \mathbf{P}^5$ and its variety of lines $X = F(Y)$ in $\mathcal{M}_6^{(2)}$

		1								
		0		0						
		0		1		0				
	0	0		0		0		0		
0		1		21		1		1		0
0	0	0		0		0		0		
	0		1			0				
		0		0						
				1						

				1						
				0		0				
			1		21	1				
	0		0		0		0			
1		21	232		21			1		
	0		0		0		0			
		1		21		1				
			0		0					
					1					

Gushel–Mukai fourfold Y and its associated double EPW sextic X in $\mathcal{M}_2^{(1)}$

		1								
		0		0						
		0		1		0				
	0	0		0		0		0		
0		1		22		1		1		0
0	0	0		0		0		0		
	0		1			0				
		0		0						
				1						

				1						
				0		0				
			1		21	1				
	0		0		0		0			
1		21	232		21			1		
	0		0		0		0			
		1		21		1				
			0		0					
					1					

“Fano variety of K3 type $\iff$ hyperkähler”?

In the case of a singular hypersurface $Y = V(f)$, we have two *mixed Hodge structures* that we can associate with Y .

- A mixed Hodge structure on the (singular) cohomology of Y .
- A (limit) mixed Hodge structure on the cohomology of a smooth hypersurface Y_∞ via a one-parameter degeneration $V(f + t \cdot g)$.

Middle H^{13} of a smooth cubic in $\mathbf{P}^{14}$, and the limit mixed Hodge structure $H_{\lim}^{13}$ of the family $V(\textcolor{red}{f}_3 + t \cdot \textcolor{blue}{g})$

0 0 0 0 1 455 5005 5005 455 1 0 0 0 0

1 20 211 20 1
 $\textcolor{blue}{434}$ $\textcolor{blue}{4774}$ $\textcolor{blue}{4774}$ $\textcolor{blue}{434}$
 1 20 211 20 1

Hodge diamonds of $(X, H) \in \mathcal{M}_6^{(1)}$ with a Coble type cubic $Y = V(\textcolor{red}{f}_3)$ in $\mathbf{P}^{14}$, and a general cubic section $Z = X \cap V(\textcolor{blue}{g}) \in |3H|$ on X

1
 0 0
 1 21 1
 0 0 0 0
 $\textcolor{red}{1}$ $\textcolor{red}{21}$ $\textcolor{red}{232}$ $\textcolor{red}{21}$ $\textcolor{red}{1}$
 0 0 0 0
 1 21 1
 0 0
 1

1
 0 0
 1 21 1
 $\textcolor{blue}{434}$ $\textcolor{blue}{4774}$ $\textcolor{blue}{4774}$ $\textcolor{blue}{434}$
 1 21 1
 0 0
 1

Expectation. The mixed Hodge structure on Y is “of $\text{K3}^{[2]}$ -type”.

Middle H^9 of the double cover of $\mathbf{P}^9$ branched along a smooth quartic, and the limit mixed Hodge structure $H_{\lim}^9$ of the family $V(f_4 + t \cdot g)$

$$\begin{array}{cccccccc} 0 & 0 & 1 & 615 & 6765 & 6765 & 615 & 1 & 0 & 0 \end{array}$$

$$\begin{array}{cccccc} 1 & 20 & 211 & 20 & 1 \\ 594 & 6534 & 6534 & 594 \\ 1 & 20 & 211 & 20 & 1 \end{array}$$

Hodge diamonds of $(X, H) \in \mathcal{M}_4^{(1)}$ with a Coble type quartic $Y = V(f_4)$ in $\mathbf{P}^9$, and a general quartic section $Z = X \cap V(g) \in |4H|$ on X

$$\begin{array}{ccccc} & & 1 & & \\ & 0 & & 0 & \\ & 1 & 21 & 1 & \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 21 & 232 & 21 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ & 1 & 21 & 1 & \\ & 0 & 0 & & \\ & & 1 & & \end{array}$$

$$\begin{array}{ccccc} & & 1 & & \\ & 0 & & 0 & \\ & 1 & 21 & 1 & \\ 594 & 6534 & 6534 & 594 \\ & 1 & 21 & 1 & \\ & 0 & 0 & & \\ & & 1 & & \end{array}$$

Expectation. The mixed Hodge structure on the double cover of $\mathbf{P}^9$ branched along Y is “of $K3^{[2]}$ -type”.